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Depletion of Common Property Resources in India: Micro-Level Evidence

N. S. JODHA

A WELL-KNOWN CONSEQUENCE OF RAPID POPULATION GROWTH in developing countries is increased pressure on land, leading to over-exploitation and degradation of the natural resource base of agriculture. The problem is greater in the low-productivity, high-risk environments such as dry tropical regions. In these regions not only is nature's bio-mass productivity low, but recovery speed of once-damaged land and vegetative resources is slow. Hence, in the face of rising population pressure, these areas run a high risk of permanent ecological degradation (UNCOD, 1977). Furthermore, these regions are among the poorest in the Third World (World Bank, 1975), and most of their inhabitants subsist on traditional agriculture. The social and economic consequences of population growth in these regions are, therefore, primarily manifested through strains and stresses faced by traditional farming systems, including the management patterns of the fragile resource base sustaining them. This essay illustrates the situation with the help of a case study of conditions in India. The focus is on rural common property resources, which have historically been an important source of collective sustenance of the people, especially the rural poor. Using micro-level data from various dry tropical parts of India, the essay narrates the process of shrinkage and degradation of common property resources.

Statement of the problem

Rural common property resources (CPRs) are broadly defined as resources to which all members of an identifiable community have inalienable use rights. In the Indian context CPRs include community pastures, community forests, waste lands, common dumping and threshing grounds, watershed drainages, village ponds, and rivers and rivulets as well as their banks and beds. The first three resources are particularly important because of their large area and their contributions to people's sustenance in dry regions. Historically, the maintenance of

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these resources reflected people's concern for their collective sustenance and for protection of fragile land resources by keeping them under grass, bush, or trees.[1] CPRs often helped generate multiple earning options and the flexibility required by farmers to adjust their farming systems to variable agroclimatic conditions (Jodha, 1986b). The changing situation of CPRs in the dry tropics must be viewed in the context of traditional farming systems in these regions.

Traditional farming systems in dry tropical regions of the developing world are currently passing through a transitional phase (McCown et al., 1979; Sandford, 1983; Livingstone, 1984; Jodha, 1986b). Having evolved during times of low population pressure and absence of rapid technical and institutional interventions, such systems were land-extensive.[2]

In the face of increased population pressure, increased integration of agriculture into the monetized market economy, and a variety of external interventions, however, traditional resource-extensive (i.e., land-extensive) farming is becoming less feasible. As elsewhere in agriculture (Boserup, 1965), the strong trend is toward increased intensity of use of land resources. This is reflected in ploughing of low-productivity submarginal lands, reduced duration and frequency of fallow, and emphasis on short duration or multiple cropping. In India, community pastures, forests, and waste lands—the major focus of extensive resource use in the past—now bear the burden of land use intensification. It happens in several ways. First, CPRs are privatized and converted into crop lands irrespective of their use capabilities (Jodha, 1986a). Second, the use intensity of CPRs, even when they are not ploughed, is increased. The latter happens because of shrinkage of CPRs and reduced availability of periodically fallowed crop lands. The final consequence is physical degradation of CPRs.

The depletion of common property resources involves the operation of several factors, jointly and individually. They can be grouped under (1) demographic factors, (2) market factors, (3) technological factors, and (4) public interventions. The literature on the subject covers different dimensions of these factors (Hardin, 1968; Ciriacy-Wantrup and Bishop, 1975; Boserup, 1965; Runge, 1981; Livingstone, 1984; Norgaard, 1984; Bromley and Chapagain, 1984; Singh, 1986; Ostrom, 1986). Figure 1 sketches the role of these factors in depletion of CPRs in the Indian context. The focal points in the process are privatization and over-exploitation of CPRs.

According to Figure 1, demographic factors influence CPRs directly by raising demand for land and indirectly by leading to privatization of CPRs. Rapid population change in association with such external factors as public interventions and commercialization induces structural changes in the village communities—changes in occupational patterns, group dynamics, and distribution of economic and political power at the local level (Mandelbaum, 1972). These changes in turn foster disregard of conventions and social sanctions that previously helped to protect CPRs.

With respect to market factors, the increased marketability and profitability of products of CPRs or of CPR-based enterprises, such as livestock rearing,

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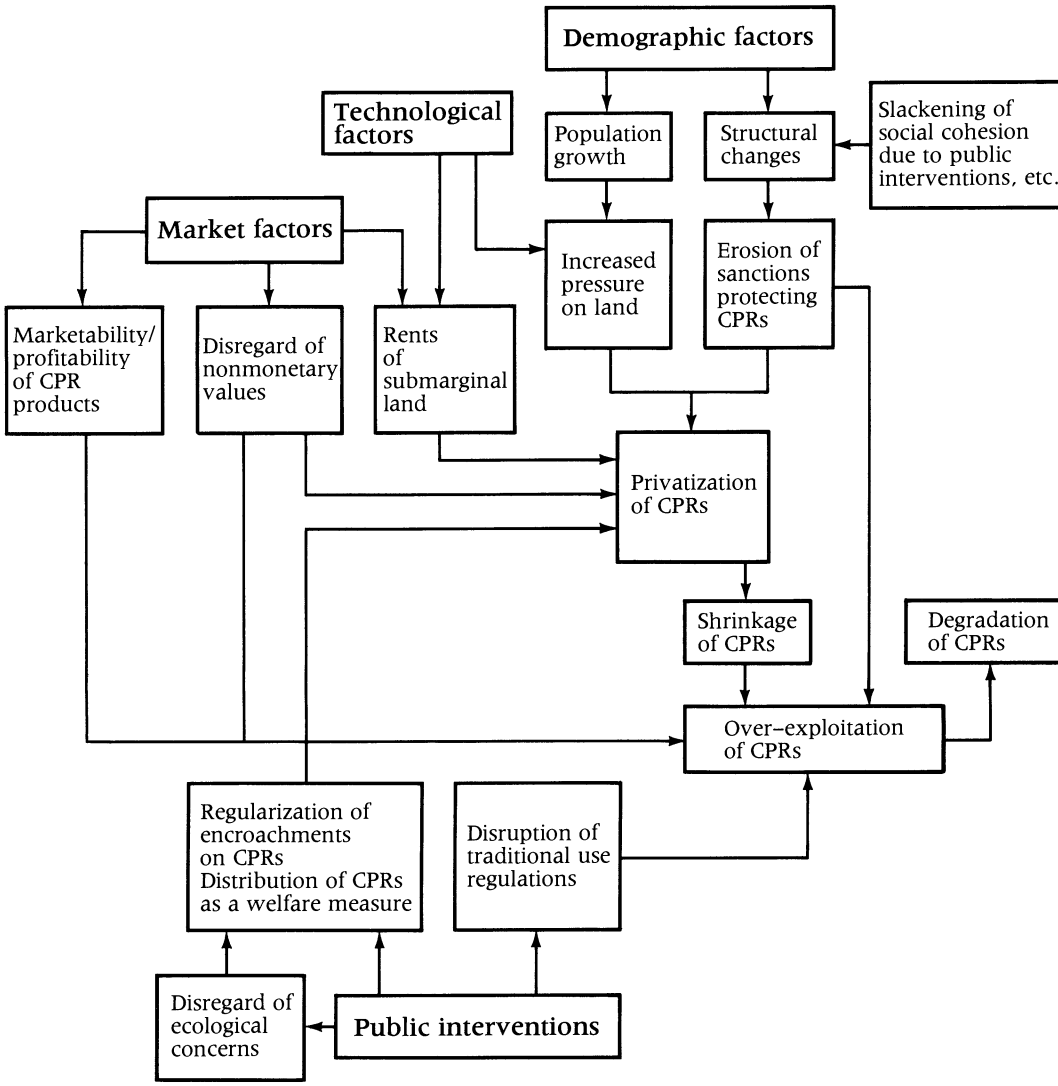
FIGURE 1 Role of various factors in the depletion of common property resources in the dry regions of India



lead directly to over-exploitation of CPRs (Jodha, 1985b). Indirectly, market forces tend to encourage individual initiative and generate disregard for collective sanctions against encroachment on CPRs (Jodha, 1985a). Finally, market forces help improve the property value or rent of submarginal lands historically kept as CPRs. This is turn encourages privatization.

Technological changes such as the introduction of irrigation and tractors also contribute to increased rent on submarginal lands and resulting shrinkage of CPRs (Jodha, 1985b).

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Public interventions including institutional reforms, welfare measures, and production-promoting activities can have adverse effects on CPRs. They tend to accentuate the roles of other factors mentioned above in depleting CPRs. For instance, land reforms in India encouraged privatization and shrinkage of CPRs. By disrupting the traditional management systems, they also contributed to over-exploitation, poor protection, and degradation of CPRs. This will be elaborated later.

Data

There is not sufficient empirical evidence on depletion of CPRs in India to fully corroborate the process sketched in Figure 1. The present essay illustrates only some aspects of the process. The evidence presented here is based on data from 82 villages in seven states in dry tropical regions of India. The fieldwork for the study was conducted during 1982–84.[3] Although the villages were widely scattered in the dry regions (see Figure 2), their selection was not random. Village selection was conditioned by the availability of suitable interviewers acquainted with the oral history of the villages and, unlike outside investigators, able to collect information on sensitive issues. Besides collection of village- and farm-level data on current status and use of CPRs, the fieldwork involved physical verification and monitoring of different aspects of CPRs. Reconstruction of the history of CPRs in each area, through oral histories and written records, was also attempted.

CPRs as sources of sustenance

Before we discuss the extent and process of depletion of common property resources in the dry regions of India, it is appropriate to briefly describe the role of CPRs in people's sustenance. CPRs contribute to employment, income, and asset accumulation. They are an important component of people's self-provisioning systems (Blaikie et al., 1985; Gupta, 1985). Because they are taken for granted, however, these contributions are seldom recognized and quantified. Some quantitative details collected as part of the author's study (Jodha, 1986a) are summarized in Table 1. Owing to their degradation and low productivity, the CPRs do not offer high returns to their users. Hence, the rural poor, who have few alternative sources of income, depend more on the low payoff options offered by CPRs. The rural rich, that is, the large landholders, depend very little on CPRs. The intermediate categories of households, not included in Table 1, depend more on CPRs than the rich.

Table 1 suggests that the rural poor receive the bulk of their fuel supplies and fodder from CPRs. CPR product collection is an important source of employment and income, especially during the periods when other opportunities are nonexistent. In certain areas employment on CPRs exceeds employ-

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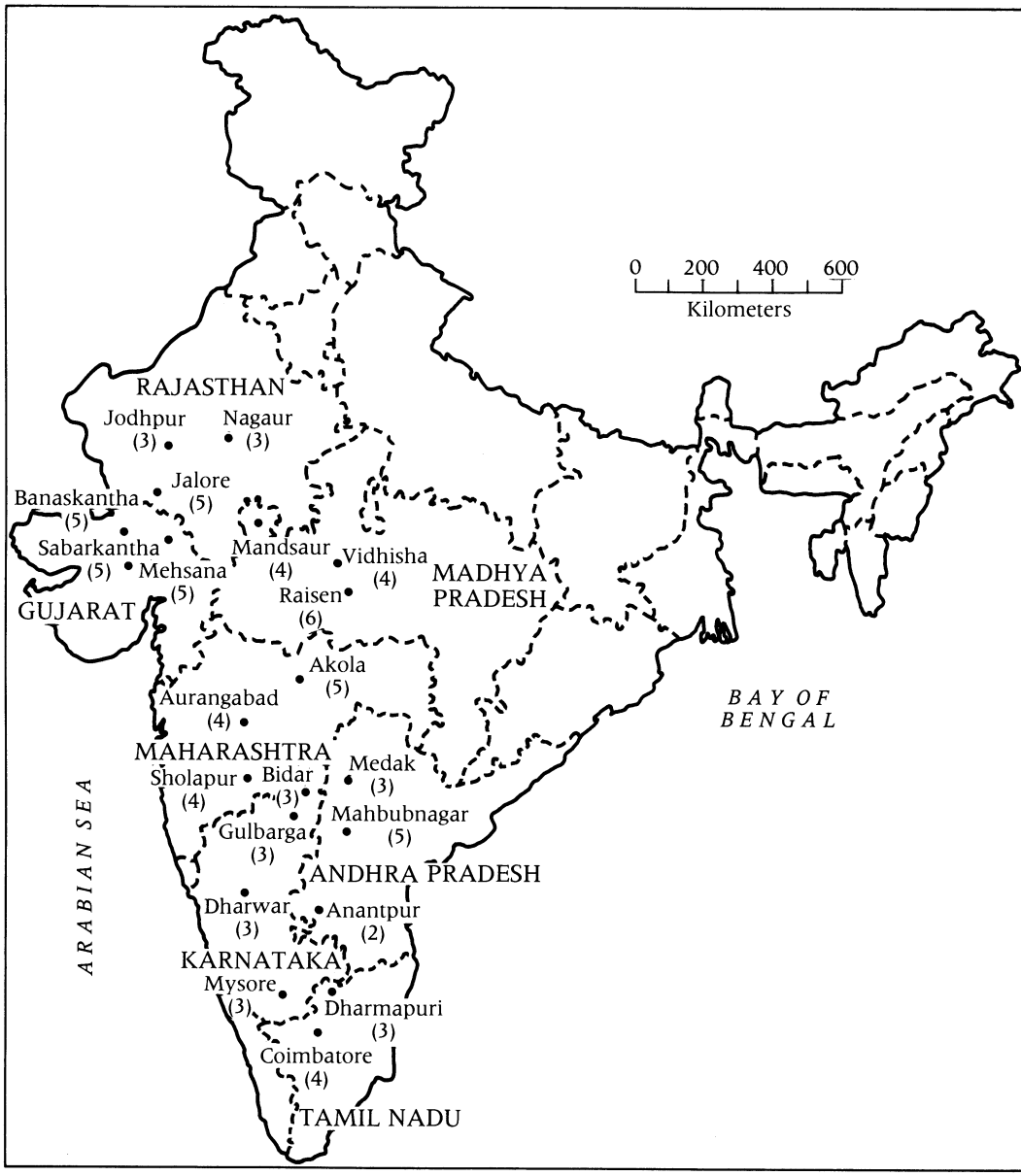
FIGURE 2 Districts and number of villages covered by the study of common property resources in India



ment generated through antipoverty public programs (Jodha, 1986a). Furthermore, CPR income accounts for 14 to 23 percent of household income from all other sources in the study villages—and it is likely this is a significant underestimate. More importantly, the inclusion of CPR incomes in total household

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TABLE 1 Extent of people's dependence on common property resources (CPRs) in dry regions of India

State (with no. of districts and villages)	Household category[a]	CPR contribution to household supplies, employment, and income: Fuel supplies[b] (percent)	Animal grazing[c] (percent)	Per household: Employment days[d] (no.)	Per household: Annual income[e] (Rs)	CPR income as percent of total income (percent)	Gini coefficient[f] for income from: All sources	Gini coefficient[f] for income from: All sources excluding CPRs
Andhra Pradesh (1, 2)	Poor	84	—	139	534	17	0.41	0.50
	Rich	13	—	35	62	1	0.41	0.50
Gujarat (2, 4)	Poor	66	82	196	774	18	0.33	0.45
	Rich	8	14	80	185	1	0.33	0.45
Karnataka (1, 2)	Poor	—	83	185	649	20		
	Rich	—	29	34	170	3		
Madhya Pradesh (2, 4)	Poor	74	79	183	733	22	0.34	0.44
	Rich	32	34	52	386	2	0.34	0.44
Maharashtra (3, 6)	Poor	75	69	128	557	14	0.40	0.48
	Rich	12	27	43	177	1	0.40	0.48
Rajasthan (2, 4)	Poor	71	84	165	770	23		
	Rich	23	38	61	413	2		
Tamil Nadu (1, 2)	Poor	—	—	137	738	22		
	Rich	—	—	31	164	2		

NOTE: This and all other tables in the essay are based on village/household data from study villages reported by Jodha (1986a).

[a]The number of sample households per village varied from 20 to 36 in different districts. The "poor" include agricultural laborers and small farm (<2 ha dryland equivalent) households. The "rich" include large farm households only.

[b]Fuel gathered from CPRs as proportion of total fuel used during three seasons covering the whole year.

[c]Animal-unit grazing days on CPRs as proportion of total animal-unit grazing days.

[d]Total employment through CPR product collection.

[e]Income mainly through CPR product collection. The estimation procedure underestimated the actual income derived from CPRs (see Jodha, 1986a).

[f]Higher value of Gini coefficient indicates higher degree of income inequalities. Calculations are based on income data for 1983–84 from a panel of households covered under ICRISAT's village-level studies. The panel of 40 households from each village included 10 households from each of the categories, namely large, medium, and small farm households and labor households.

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incomes from all other sources reduced the extent of rural income inequality, as indicated by lower values of the Gini coefficient. Despite their valuable contributions to the economy of the rural poor, CPRs are one of the most neglected areas in development planning in the country. The consequence is their depletion both in terms of decline in area and in terms of physical degradation.

Depletion of CPRs

Of the two forms of depletion of CPRs, decline in area is more easily captured by written and oral records of village land use. In contrast, the extent of physical degradation of CPRs, though quite visible and keenly felt by villagers, is difficult to quantify because their productivity in the past was not recorded.

The decline in CPR area in the study villages is reported in Table 2. The information relates to 82 villages from 21 districts from dry regions of seven states in India.[4] The benchmark is the period 1950–52, which was a watershed in the agrarian history of independent India. During the early 1950s, comprehensive land reforms were introduced. The status and management pattern of CPRs underwent drastic changes following land reforms and subsequent public interventions. Because they were a major event in the memory of villagers, the land reforms also served as an important reference point to facilitate recording an oral history of CPRs. The benchmark situation was compared with the status of CPRs during 1982–84, when fieldwork for the study was conducted.

According to Table 2, the area of CPRs declined by 31 to 55 percent in various states. The impact of this change is clearly visible in terms of both decline in proportions of CPR lands in total village area and increase in population pressure on CPRs.

In the absence of recorded information with which to assess degradation or decline in productivity of CPRs over time, a benchmark had to be constructed from oral histories and scattered village records. The comparison of this benchmark with the present situation, shown in Table 3, gives some indication of the qualitative decline of CPRs.

The first indicator of physical degradation of CPRs is the almost complete disappearance of a number of products that villagers used to gather from the commons before the 1950s. According to Table 3, not even half the number of items gathered in the past were available in the early 1980s in any of the study villages.

The variety of CPR products declined because a number of plant and tree species disappeared from common lands. Local names of a limited number of CPR plots indicated their coverage by specific vegetation in the past. Few of those species grow there any longer. Similarly, a number of selected CPR plots that were traditionally used for grazing more productive animals like lactating cattle and working bullocks are no longer able to sustain such animals. With their

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TABLE 2 Decline of area of CPR land in dry regions of India

State (and no. of districts)	Number of study villages	Area of CPRs[a] 1982–84 (ha)	CPRs as proportion of total village area		Decline in area of CPRs 1950–52 to 1982–84 (percent)	Persons per 10 ha of CPR area	
			1982–84 (percent)	1950–52 (percent)		1951	1981
Andhra Pradesh (3)	10	827	11	18	42	48	134
Gujarat (3)	15	589	11	19	44	82	238
Karnataka (4)	12	1165	12	20	40	46	117
Madhya Pradesh (3)	14	1435	24	41	41	14	47
Maharashtra (3)	13	918	15	22	31	40	88
Rajasthan (3)	11	1849	16	36	55	13	50
Tamil Nadu (2)	7	412	10	21	50	101	286

NOTE: Table adapted from Jodha (1986a), where more disaggregated details are reported.
[a]CPRs include community pasture, village forest, waste land, watershed drainage, river and rivulet banks, and other common lands. Data indicate average area per village.

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TABLE 3 Physical degradation of CPRs in dry regions of India

Indicator of changed status	State (with no. of villages)						
	Andhra Pradesh (3)	Gujarat (4)	Karnataka (2)	Madhya Pradesh (4)	Maharashtra (3)	Rajasthan (4)	Tamil Nadu (2)
Number of CPR products collected by villagers[a]							
In the past	32	35	40	46	30	27	29
At present	9	11	19	22	10	13	8
Number of trees and shrubs per hectare in							
Protected CPRs[b]	476	684	662	882	454	517	398
Unprotected CPRs	195	103	202	215	77	96	83
Number of watering points (ponds) in grazing CPRs							
In the past	17	29	20	16	9	48	14
At present	4	13	4	3	4	11	3
Number of CPR plots where rich vegetation, indicated by their nomenclature, is no longer available	—	12	3	6	4	15	—
CPR area used for cattle grazing in the past, currently grazed mainly by sheep/goats (ha)[c]	48	112	95	—	52	175	64

NOTE: Based on observation and physical verification of current status (during 1982–84) and on past details collected from oral and recoded description of CPRs in different villages. The choice of CPRs where plot-based data are reported was guided by availability of past information about them.

[a]Includes types of fruits, flowers, leaves, roots, timber, fuel, and fodder in the villages. The "past" indicates the period preceding the 1950s; the "present" indicates the early 1980s.

[b]Protected CPRs were the areas (called Oran) where for religious reasons live trees and shrubs are not cut. The situation of CPR plots (numbering two to four in different areas) was compared with other bordering plots of CPRs that were not protected by religious or other sanctions.

[c]Relates to area covered by specific plots, traditionally used for grazing high-productivity animals (e.g., cattle in milk, working bullocks, or horses of feudal landlords). Because of the depletion of such plots, such animals are no longer grazed there.

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depleted forage potential and changed vegetative composition, they are now grazed by sheep and goats instead of cattle.

The difference in the number of species found on protected and unprotected CPRs is an important indicator of vegetative degradation and associated resource depletion. Certain CPR areas are protected through religious sanctions against removal of live trees and shrubs from them. The number of trees and shrubs recorded per hectare was three to six times higher on protected CPRs than on unprotected ones. The number of watering points, an important component of common grazing lands, has also declined (Table 3).

The process of change

Among factors contributing to the depletion of CPRs, we consider first privatization due to public policies. The inter-village differences in the status of CPRs caused by demographic and other factors are discussed later. In the following discussion, we do not address separately the decline of CPR area and the physical degradation of CPRs. We proceed with the premise that shrinkage and unregulated use of CPRs inevitably lead to their physical degradation.

Public policy and decline of CPR area

Public policies that followed the introduction of land reforms in the early 1950s encouraged privatization of CPRs and led to a rapid decline in their area. Although public policies did not explicitly advocate privatization, in practice virtually all the programs designed to provide land to the landless and to small landholders have reduced the area of CPRs, because little other land for redistribution could be acquired through land ceiling laws (Ladejinky, 1972) or through voluntary donations.[5] Privatization involved distribution of land to the landless and the rural poor under various welfare schemes, and legalization of encroachment on CPR lands.[6] Although privatization was supposed to benefit the rural poor, in practice, as will be shown later, the benefits to relatively better-off households exceeded the benefits to the poor. Similarly, despite the mention of rehabilitation and protection of grazing lands and village forests in the government's planning documents, very little was done in practice (Jodha, 1986a). Moreover, development programs for these resources were highly "technique"-dominated, with little attention given to user perspectives.

Further, such institutional changes in village administration as the abolition of the feudal system, the decline in the authority of the traditional village headman (Mandelbaum, 1972), and the introduction of formally elected village councils (*panchayats*) led to slackening of the traditional arrangements for conservation and regulated use of CPRs (Jodha, 1985b).

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Privatization of CPR lands

Table 4 depicts the distribution of privatized CPR lands in the study villages. The proportion of poor households among the total recipients of CPR lands was generally the highest; however, the proportion of land received by the poor was much lower than that received by other social groups. The average land area received by the poor barely exceeded one hectare per household; the area received by "others" ranged from 2 to 3 hectares. The last column of Table 4 reveals that those who already had more land also received more privatized land.

A detailed inquiry into the current status of CPR lands received by the rural poor revealed that the poor could not sustain the gains from land distribution programs. According to Table 5, if Tamil Nadu is excluded, 19 to 42 percent of the land-receiving poor households disposed of part or all of the land received, through sale, mortgage, and so on. This happened despite legal restrictions in several areas on transactions in land belonging to the poor. The area so disposed amounted to 23 to 45 percent of the land received in different states, excluding Tamil Nadu. Nine to 36 percent of lands so disposed were submarginal and unsuitable for cropping. Most of the households that disposed of land did not have the necessary resources—like bullocks and ploughs—to cultivate it. Lack of resources to develop the land, inadequacy of the land to sustain the family, and pressing needs for cash induced the poor households to give up the land.

The evidence suggests that policies and programs to help the rural poor through privatization of CPRs have not worked as intended. It is quite doubtful whether poor people's collective loss from reduced CPR areas has been compensated by private ownership of CPR lands given to them.

Slackening of use regulations

The loss to the users of common property resources has resulted not only from reduced area, but also from degradation or reduced productivity of CPRs. The latter is partly an indirect consequence of shrinkage and overcrowding on CPRs. More importantly, as a byproduct of public interventions, including land reform and introduction of democratic institutions at the village level, the traditional formal or informal arrangements regulating the use of CPRs have weakened or disappeared (Jodha, 1985b). For instance, the abolition of grazing taxes and levies and of compulsory labor to fence and trench CPR lands and to desilt ponds has led to significant declines in the private costs to CPR users. Similarly, the replacement of the informal authority of the village elders by elected village councils and the replacement of conventions by formal laws relating to land have eroded the traditional social and religious sanctions protecting CPRs. This in turn has led to over-exploitation and poor upkeep of CPRs. Table 6 indicates the

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TABLE 4 Distribution of privatized CPR lands to different household groups in dry regions of India

State (with no. of villages)	Total land privatized (ha) (1)	Total number of households receiving land (2)	Share of poor[a] in (1) (percent) (3)	Share of poor in (2) (percent) (4)	Land per household received by		Average land size after receiving new land	
					Poor[a] (ha) (5)	Others[b] (ha) (6)	Poor[a] (ha) (7)	Others[b] (ha) (8)
Andhra Pradesh (6)	493	401	50	74	1.0	2.1	1.6	5.0
Gujarat (8)	287	166	20	45	1.0	2.6	1.8	9.4
Karnataka (9)	362	203	43	65	1.3	3.0	2.2	8.0
Madhya Pradesh (10)	358	204	42	62	1.2	3.2	2.5	9.5
Maharashtra (8)	316	227	38	53	1.1	1.9	2.0	6.2
Rajasthan (7)	635	426	22	36	1.2	3.2	1.9	7.2
Tamil Nadu (7)	447	272	49	66	1.0	1.5	1.9	6.7

NOTE: Table adapted from Jodha (1986a). Three districts were covered in each state except Andhra Pradesh and Tamil Nadu, where two districts each were covered.
[a]The "poor" include agricultural laborers and small farm (<2 ha of dry land equivalent) households.
[b]"Others" include both medium and large farm households.

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TABLE 5 Dispossession of privatized CPR lands by rural poor in dry regions of India

State[a]	Dispossessed land: As proportion of total land received (percent)	Dispossessed land: Containing area unsuited to cropping (percent)	Dispossessing households as proportion of total land recipients (percent)	Distribution of households by reasons for land dispossession: No bullocks, plough (percent)	Inability to develop land (percent)	Land too small, poor (percent)	Pressing need for cash (percent)
Andhra Pradesh	44	25	30	81	14	15	16
Gujarat	45	15	29	53	23	17	8
Karnataka	32	24	42	72	30	34	12
Madhya Pradesh	23	9	19	47	17	12	2
Maharashtra	40	16	36	53	27	21	9
Rajasthan	33	36	27	52	25	17	2
Tamil Nadu	17	24	18	77	18	9	—

NOTE: Table adapted from Jodha (1986a). For definition of "Poor" see Table 4, note a.
[a]For number of districts and villages covered see note to Table 4.

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TABLE 6 Changes in management of CPRs in dry regions of India

State (with no. of villages)	Number of villages pursuing the following measures					
	Formal/informal regulations on CPR use[a]		Formal/informal taxes/levies on CPR use[c]		Users' formal/informal obligation toward upkeep of CPRs	
	In the past[b]	At present[b]	In the past	At present	In the past	At present
Andhra Pradesh (10)	10	—	7	—	8	—
Gujarat (15)	15	2	8	—	11	2
Karnataka (12)	12	2	9	—	12	3
Madhya Pradesh (14)	14	2	10	—	14	3
Maharashtra (13)	11	1	6	—	10	1
Rajasthan (11)	11	1	11	—	11	2
Tamil Nadu (7)	7	—	4	—	7	1

NOTE: Table based on fieldwork during 1982–84 (Jodha, 1986a).
[a]Measures such as regulated/rotational grazing, seasonal restrictions on use of CPRs, provision of CPR watchmen.
[b]The "past" indicates the period prior to the 1950s; the "present" indicates the early 1980s.
[c]Measures such as grazing taxes and penalties for violation of regulations on use of CPRs. See Jodha (1985b) for descriptive account.
[d]Measures such as contribution toward desilting of watering points, fencing, and trenching.
— indicates nil.

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extent of change in terms of discontinuation of several practices that once constituted integral parts of formal or informal management systems for CPRs. According to the table, in most cases not even a tenth of the villages that formerly followed various management practices with respect to CPRs do so at present. The consequences of this change hardly need elaboration.

The role of other factors

Public policies and programs encouraging privatization of CPRs affected all villages. But the actual decline in CPRs differed from village to village even within the same state or district. Table 7 shows the range of inter-village differences in the extent and decline of CPRs. These differences call for examination of other factors that might influence CPRs independently or in association with public policies.

These factors can be grouped into three categories. Population size or its density, growth of population, and village size in terms of number of households represent the demographic factors. Second, the ecological factors comprise such variables as area of the village, extent of submarginal lands,[7] and number of livestock. Finally, the role of market forces is reflected in such variables as distance from a marketing center.

Table 8 presents a comparison of groups of villages with the highest and lowest area of CPRs along with the values for factors influencing the status of

TABLE 7 Inter-village differences in CPRs as a proportion of village area and decline in area of CPRs

State (with no. of villages)	CPR area as percent of total village area		Percent decline in CPR area, 1950–52 to 1982–84
	1950–52	1982–84	
Andhra Pradesh (10)	9–30	5–20	26–56
Gujarat (15)	7–31	2–23	21–69
Karnataka (12)	6–36	4–30	16–50
Madhya Pradesh (14)	29–69	19–47	14–51
Maharashtra (13)	8–43	6–34	14–52
Rajasthan (11)	20–49	8–26	17–71
Tamil Nadu (7)	7–39	5–23	21–65

NOTE: Table based on fieldwork and records (Jodha, 1986a).

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TABLE 8 Inter-village difference in the extent of CPRs and possible factors responsible for them, 1950–52

		Average value of variable for each group of villages							
State	Village group[a] (1)	CPRs as proportion of col. (3) (2)	Total area (ha) (3)	Size (and density[b]) (4)		Number of households (5)	Submarginal lands as proportion of col. (3)[c] (percent) (6)	Livestock per 10 ha of (3)[d] (no.) (7)	Distance from market center[d] (km) (8)
Andhra Pradesh	Bottom	13	1134	1036	(91)	194	35	12	7
	Top	25	648	537	(83)	103	43	11	13
Gujarat	Bottom	14	588	895	(152)	192	12	12	8
	Top	27	243	369	(149)	78	32	18	10
Karnataka	Bottom	12	1110	1013	(91)	157	22	8	14
	Top	31	624	529	(85)	97	38	10	19
Madhya Pradesh	Bottom	35	593	293	(49)	56	36	7	14
	Top	52	218	183	(84)	35	49	13	18
Maharashtra	Bottom	15	632	545	(86)	82	13	12	9
	Top	38	246	250	(101)	50	31	15	21
Rajasthan	Bottom	22	642	358	(56)	75	55	12	13
	Top	46	1164	551	(48)	98	60	12	15
Tamil Nadu	Bottom	10	734	1242	(169)	252	16	18	9
	Top	37	390	1343	(346)	198	52	40	13
Correlation coefficient		1.0	.33	−.51		−.56	.56	.16	.57

NOTE: Table based on fieldwork and records (Jodha, 1986a). The table covers 56 out of 82 villages.

[a]Villages in each state were arranged in descending order of the proportion of CPR area to total village area during 1950–52. One-third of the villages at the top end and the bottom end of the list for each state were picked to form two groups, described as the "bottom" and "top" groups of villages accordingly. The average situation of each of the two groups with reference to different variables is presented.

[b]Figures in parentheses indicate number of person/km^2.

[c]Lands as defined in note 7.

[d]Information on livestock and distance from market center relates to 1982–84 because the details for 1950–52 could not be collected for all villages. However, this does not significantly influence the results.

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CPRs. Table 9 compares villages with the highest and lowest declines in CPRs over time. The important generalizations emerging from these comparisons are summarized in Table 10.

Table 10 suggests a negative association between the extent of CPRs and such factors as size of population and number of households in the village. This is not difficult to understand. Historically, even in the context of a micro unit like a village, higher population pressure implied land scarcity. Once the supply of better lands had been exhausted, any additional demand could be met only through occupation of submarginal lands, including CPRs. This process was intensified with the rapid increase in population over a relatively short period from the early 1950s to the early 1980s. Increased population pressure, as mentioned earlier, induces a shift from land-extensive to land-intensive farming. CPRs bear the maximum burden of land use intensification.

As observed during the fieldwork, smaller villages were relatively isolated from major markets. They were located in ecologically less favorable environments with greater proportions of submarginal lands. They were also found to have greater social cohesion and respect for conventions relating to use of CPRs. These attributes of smaller villages, originating in ecological, social, demographic, and market-related factors, helped protect the CPRs. This explains both the greater initial availability of CPRs and the lower decline in their area over time in smaller villages.

Table 10 indicates a positive association between the initial extent of CPRs and village area, extent of submarginal lands, and livestock density. Submarginal lands and livestock rearing have a long historical association. Viewed in the context of the traditional practice of reserving submarginal lands for common use, this explains the positive association between the importance of livestock and the extent of CPRs. Furthermore, in the villages with higher livestock populations, people had greater stake in CPRs—a factor that also slowed their decline since the early 1950s (Table 10).

A positive association between the decline in CPRs and the extent of submarginal lands as well as the initial area of CPRs (Table 10) can be explained with reference to public policies, which encouraged privatization of CPRs as discussed earlier. Villages with higher initial extent of CPRs lost more of CPR area because it was easier to execute land distribution programs in them. In addition, villagers had a greater awareness of the status of CPRs where the total CPR area was smaller. This prevented encroachments on CPRs by fellow villagers as well as by the state. This sort of safeguard against the decline of CPRs was rarely found in the villages with large and widely scattered CPR lands.

Table 10 indicates a positive association between the extent of CPRs and distance from a market center. It also shows a negative association between the decline in CPR area and distance from a market center. Nearness to a market center implies emergence of individualistic tendencies among the villagers that may reduce reliance on collective sustenance through CPRs.

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TABLE 9 Inter-village differences in the extent of decline of CPR area from 1950–52 to 1982–84 and possible factors responsible for them

		Average value of variable for each group of villages							
State	Village group[a]	Decline in CPR area (percent)	Increase in population 1951–81 (percent)	Number of households in 1981	Area of village (ha)	Initial extent of CPR area (1951) (ha)	Area of submarginal land[b] (ha)	Livestock per 10 ha, 1982–84 (no.)	Distance from market center, 1982–84 (km)
Andhra Pradesh	Bottom	28	66	119	448	106 (24)[c]	172 (38)[c]	15	13
	Top	53	73	277	986	143 (16)	320 (36)	12	9
Gujarat	Bottom	27	64	228	266	55 (20)	42 (16)	17	12
	Top	58	77	181	442	77 (18)	96 (22)	13	8
Karnataka	Bottom	27	49	182	583	102 (17)	113 (19)	9	16
	Top	46	52	216	1131	230 (20)	366 (32)	9	17
Madhya Pradesh	Bottom	20	74	50	261	95 (34)	87 (34)	9	19
	Top	48	72	109	706	294 (49)	348 (49)	7	13
Maharashtra	Bottom	20	57	73	289	82 (29)	81 (28)	14	19
	Top	44	65	87	346	88 (25)	64 (18)	11	14
Rajasthan	Bottom	32	67	125	642	140 (22)	355 (55)	12	15
	Top	65	90	164	1336	539 (40)	721 (54)	9	13
Tamil Nadu	Bottom	27	38	349	733	75 (10)	119 (16)	18	10
	Top	62	38	239	551	146 (26)	219 (40)	19	12
Correlation coefficient		1.0	.36	.31	.52	.50	.51	−.2	−.31

NOTE: Table based on fieldwork and records (Jodha, 1986a). The table covers 56 out of 82 villages.

[a]Villages in each state were arranged in descending order of percentage decline in their CPR area from 1950–52 to 1982–84. One-third of villages at the top and bottom ends of the list for each state were picked to form two groups, described as the "top" and "bottom" groups of villages accordingly. The average situation of each of the two groups with reference to different variables is presented.

[b]For definition of submarginal lands see note 7.

[c]Figures in parentheses indicate proportion of CPR lands and submarginal lands to total area of the vi

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TABLE 10 General inferences from the tabular analysis of village-level data indicating association between different attributes of villages and relative position of their CPRs

	Relative position of CPRs			
	Extent of CPR area (1950–52)		Decline in CPR area (1950–52 to 1982–84)	
Attribute of villages	**Top**	**Bottom**	**Top**	**Bottom**
Demographic factors				
Higher population		✓		
Greater number of households		✓	✓	
Higher population increase over time			✓	
Ecological factors				
Larger area of village	✓		✓	
Larger extent of submarginal lands	✓		✓	
Larger initial area of CPRs			✓	
Greater importance of livestock	✓			✓
Market factor				
Greater distance from market center	✓			✓

NOTE: Table based on comparison of groups of villages in each state having highest and lowest extent of CPRs (1950–52) and groups having the highest and lowest decline of CPRs (1950–52 to 1982–84). For details see Tables 8 and 9.

Regression results

The broad trends suggested above were verified with regression analysis. The data for all 82 villages were subjected to linear regression. The extent of CPR area in 1950–52 and in 1982–84 and the decline in area during this period were treated as dependent variables. They were regressed on the independent variables listed in Table 11.

We expected that the size of the CPR area would be negatively influenced by the size of the population and positively influenced by total village area, distance to market, the area of submarginal land, and livestock density. As indicated by the results reported in Table 12, these expectations were largely confirmed. The extent of submarginal area significantly influenced the extent of CPRs in both periods. Size of village population was another significant determinant of the extent of CPRs. As with most of the variables, however, the estimated effect of population on size of CPR area declined in 1982–84 compared with 1950–52. Other factors such as the importance of livestock and the distance from a market center did not emerge as significant determinants of CPR area. Moreover, these variables were less influential in explaining the variation in CPR area in the 1980s than in the 1950s.

Demographic factors declined in importance in the 1980s, while the area of submarginal lands (an ecological variable) remained the only highly signifi-

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TABLE 11 Means and standard deviations of CPR area and its determinants in 82 study villages

Variable	Unit	Mean	Standard deviation
Dependent variables			
CPRs (early 1950s)[a]	Hectare	159.79	143.50
CPRs (early 1980s)	Hectare	88.45	62.40
Change in CPRs (early 1950s to early 1980s)	Hectare	−71.34	−88.25
Independent variables			
Population (1951)	Number	571.00	381.00
Population (1981)	Number	916.00	569.00
Household (1951)	Number	110.00	73.00
Household (1981)	Number	177.00	110.00
Total village area (1951)[b]	Hectare	618.11	406.18
Submarginal area (1951)[b]	Hectare	225.78	216.92
Livestock per 10 ha (1981)	Number	13.21	5.90
Market distance (1981)	Km	13.61	6.00
Change in population from 1951 to 1981	Number	345.00	205.00

NOTE: Table based on fieldwork and records (Jodha, 1986a).
[a]This was also treated as an independent variable while regressing change in CPR area on other variables.
[b]Extent of area remained constant over time.

TABLE 12 Determinants of CPR area in 82 study villages during the early 1950s and early 1980s

Explanatory variable	CPR in 1950s	CPR in 1980s
Total area of village	0.11 ** (2.08)	0.05 (1.68)
Population	−0.08 ** (−2.63)	−0.02 * (−1.98)
Submarginal area	0.47 ** (6.32)	0.19 ** (4.65)
Livestock density (livestock/10 ha)	2.00 (1.18)	0.48 (0.53)
Market distance	−0.57 (−0.47)	−0.52 (−0.81)
Intercept	12.89 (0.36)	23.85 (1.30)
R^2	0.82	0.74
F ratio	76.01	45.85
Number of observations	82	82

NOTES: Table based on fieldwork and records (Jodha, 1986a). Figures in parentheses are t values; * and ** denote statistical significance at the .05 and .01 levels respectively.

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TABLE 13 Determinants of change in CPR area in 82 study villages from early 1950s to early 1980s

Explanatory variable	Regression coefficient
Change in population	−0.05 ** (−2.16)
CPR in 1950	−0.58 ** (−29.47)
Household number in 1950	−0.11 (−1.76)
Intercept	−25.86 ** (−4.93)
R^2	0.94
Number of observations	82

NOTES: Table based on fieldwork and records (Jodha, 1986a). Figures in parentheses are *t* values; ** denotes statistical significance at the .01 level.

cant determinant of CPRs in that decade. With the rapid privatization of CPRs since the early 1950s, the extent of CPRs in many villages had shrunk by the early 1980s to a level that was quite independent of population pressure and of such factors as distance from a market and importance of livestock. The only factor limiting the privatization of CPRs for arable farming was the submarginal quality of land. It is no wonder, therefore, that this variable remained a highly significant determinant of CPR area in the 1980s.

Table 13 presents the regression coefficients of variables influencing the decline in CPR area from the early 1950s to the early 1980s. The initial (i.e., early 1950s) extent of CPR area was the most significant factor influencing the decline in CPR area. The effect of the percentage change in village population from 1951 to 1981 was also statistically significant and in the expected direction. The number of households in 1951 did not emerge as a significant variable in the regression. This could be partly due to interregional differences in the relative size of villages, the impact of which could not be captured in the regression analysis.[8]

Conclusion

This analysis of village- and farm-level data from 82 villages demonstrates the very important role played by common property resources in the sustenance of rural people in dry tropical regions of India. In recent decades, CPRs have declined both in their area and in their productivity. Population growth leading to increased pressure on land has historically contributed to decline in CPRs, which often represented an extensive pattern of land use. This is borne out by the comparative analysis of the initial period (1950–52) in the present case. Follow-

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ing the initial period, however, the effect of rapid population growth must be assessed in conjunction with public policies, which probably would have led to decline of CPRs even in the absence of population pressure. The erosion of traditional effective CPR management systems is clearly a consequence or side effect of public interventions. An important implication of the study is that as long as public interventions are a key factor affecting CPRs, a reorientation of the former should be emphasized to rehabilitate CPRs. A restriction on further privatization of CPRs, introduction of use regulations supported by some element of user cost, and fiscal incentives to village councils should form part of CPR strategies for the future.

Notes

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1 This is partly indicated by the following traditional practices observed in a number of villages. As an indicator of concern for the community's collective needs, the area of permanent pastures (i.e., a grazing CPR) was periodically revised to match the increase in number of cattle. Similarly, to prevent intensive use of highly erodible shallow soils, ponds with vast catchments were dug. Ploughing and cutting of shrubs were prohibited in the catchments of ponds, and statues of deities were erected in such areas as warnings. Vegetation in such places, called Oran, was protected through religious sanctions. For examples of people's concern for CPRs in other countries, see Ostrom (1986) and Livingstone (1984).

2 The land-extensive character of traditional farming systems is reflected in the use of long-duration, indeterminate cultivars, seasonal fallows, crop-bush fallows, and mixed farming (Jodha, 1986b).

3 Jodha (1986a) describes the methodology and procedures used in conducting the study. The present essay at times draws on the more detailed results of the study reported elsewhere (Jodha, 1985a; 1985b; 1986a).

4 The states and districts are as follows: (1) Andhra Pradesh: Anantpur, Mahbubnagar, and Medak; (2) Gujarat: Banaskantha, Mehsana, and Sabarkantha; (3) Karnataka: Bidar, Dharwad, Gulbarga, and Mysore; (4) Madhya Pradesh: Mandsaur, Raisen, and Vidisha; (5) Maharashtra: Akola, Aurangabad, and Sholapur; (6) Rajasthan: Jalore, Jodhpur, and Nagaur; (7) Tamil Nadu: Coimbatore and Dharmapuri.

5 Under a movement called *Bhoodan*, initiated by Vinoba Bhave, the voluntary donations of surplus land by landowners for redistribution to the poor was encouraged.

6 State acquisition of CPR land for construction of public facilities like schools and roads or for reforestation is another way in which CPRs are curtailed. This is not discussed here.

7 The submarginal lands, for our purpose, include those that are rocky, gravelly, or extremely sandy, as well as those having problems like salinity, water-logging, and highly undulating topography. Traditionally such lands were kept as village common.

8 For instance the larger villages in low-population-density districts of Madhya Pradesh, Rajasthan, or Maharashtra, with their relatively lower extent and higher decline of CPR area, proved to be smaller than the smallest villages in the districts in Tamil Nadu or Andhra Pradesh having higher population densities. The impact of village size (i.e., number of households) on CPRs lost its significance when villages from all regions were pooled for regression.

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